



CedarCare

Your AI Health Guide for BC

CSIS 4495 · FINAL PRESENTATION · WINTER 2026

CedarCare

Your AI Health Guide for BC

Desmond Chua · 300369803

CSIS 4495-002 · Applied Research Project · Douglas College · Winter 2026

● cedarcare.streamlit.app

● github.com/desporamon/W26_4495_S2_DesmondC



CedarCare AI

● Online

I have chest pain and my left arm feels numb

How severe is the chest pain?

Mild

Moderate

Severe

Unbearable

Severe



These symptoms may indicate a cardiac emergency. Please call 911 immediately or go to your nearest emergency room.

Do not drive yourself.

● Emergency pathway detected · CTAS Level 1

The Problem

3.1 hrs

Average BC ER
wait time (2023)

895K

BC residents without
a family doctor

0

Canadian AI symptom
checkers using CTAS

No tool exists that translates clinical triage logic into guidance BC residents can use before entering the system.

Why this matters in BC:

- No Canadian AI symptom checker incorporates CTAS — the triage standard used in every Canadian ER
- BC-specific resources (UPCCs, HealthLink 8-1-1, MSP virtual care) not referenced by any existing tool
- Co-op at Richmond Hospital: ~15–20% of ED patients were uncertain if they needed emergency care

Most competitors triage well or guide well — none do both, for BC.



The Solution & Novelty

Three-Layer AI Pipeline

LAYER 1

GPT-4o-mini

Natural language extraction only

Converts free text → structured symptom data (JSON)



LAYER 2

CTAS Rules

Deterministic emergency safety

14 conditions — fires immediately on match, ML never runs



LAYER 3

Random Forest

ML triage classification

≥70% confidence → prediction | <70% → OpenAI fallback

What Makes CedarCare Novel

Only CTAS-integrated consumer tool

The Canadian Triage and Acuity Scale is the clinical standard in every Canadian ER. No consumer app has ever incorporated it.

BC-specific care routing

Recommendations reference UPCCs, MSP-covered virtual care, and HealthLink 8-1-1 — resources generic platforms don't know exist.

Dual ML architecture

Supervised Random Forest for individual triage. Unsupervised K-Means for population analytics. Two ML approaches, one platform.

Utility — Who Benefits and How



Make the right care decision at 2am

895K BC residents without a family doctor get structured triage guidance before entering the system — reducing unnecessary ER visits and helping people find the right level of care.

895K residents without a GP



Your health history, not just a one-off check

The platform builds a longitudinal health profile across sessions. Patterns that only emerge over repeat visits get surfaced — something no existing symptom checker does.

Personal Health Dashboard



Population insight for health planners

The system analytics dashboard gives BC Health Authorities symptom trend data and K-Means patient segmentation — a potential B2B pathway beyond the individual consumer.

B2B — BC Health Authorities

12

Survey respondents

4.33/5

Ease of use

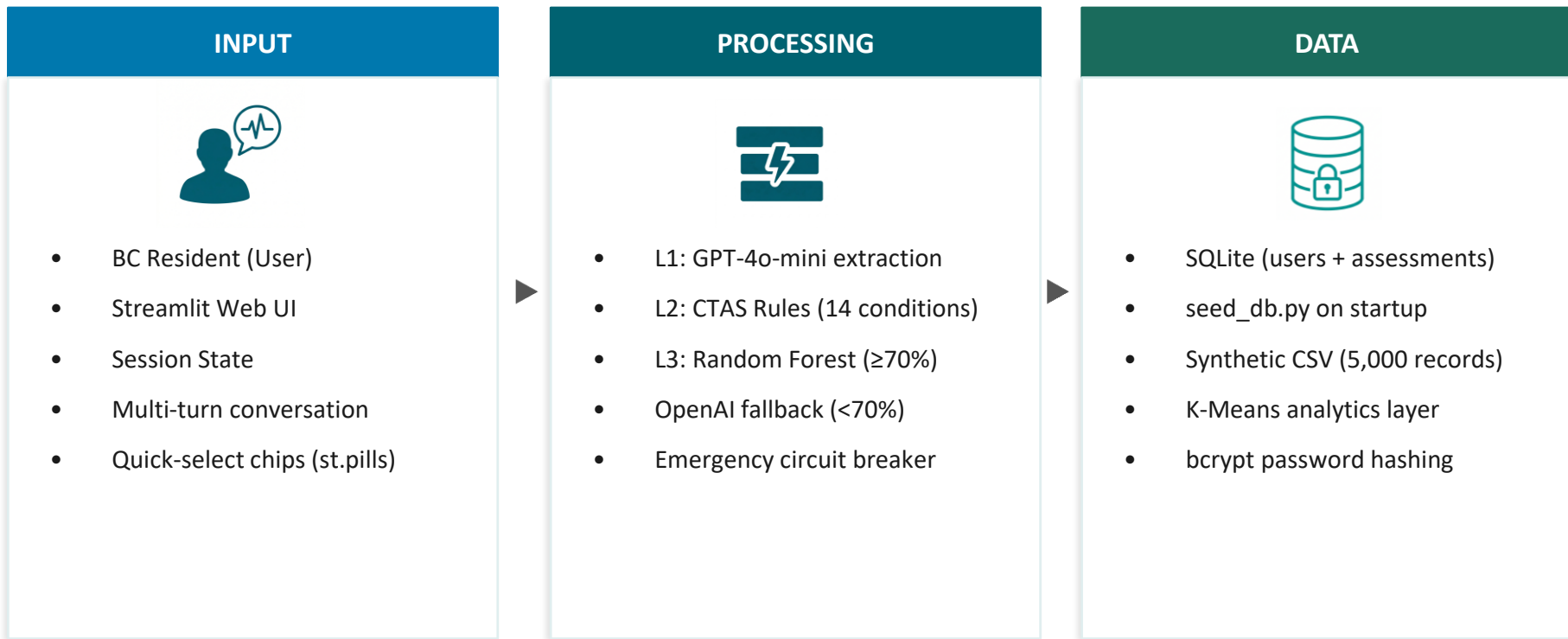
4.42/5

Likelihood to recommend

83%

Named chatbot most useful

Full System Framework



Technology Stack

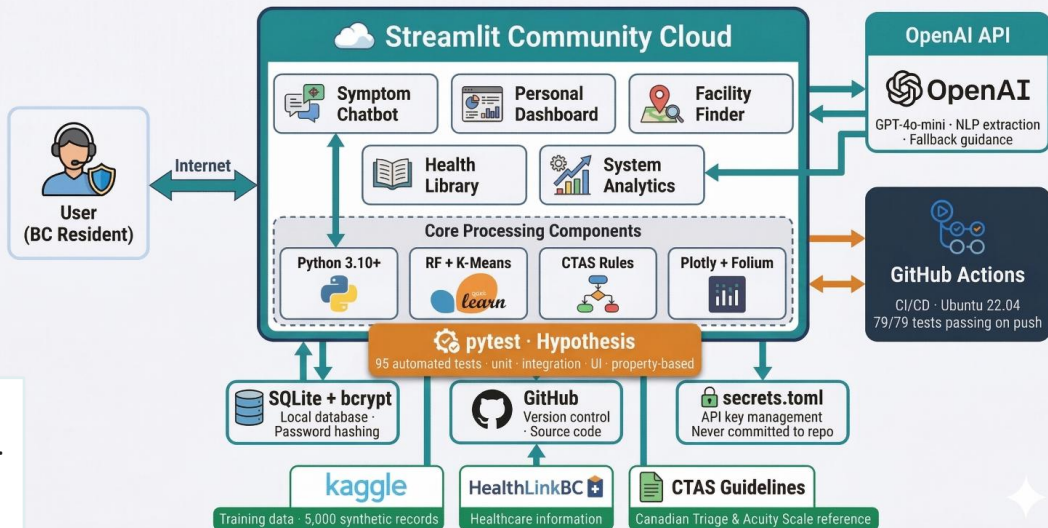
Stack by Layer

Frontend	Python · Streamlit · Folium · st.pills()
AI / ML	GPT-4o-mini · scikit-learn · joblib · pandas
Data	SQLite · bcrypt · Plotly · synthetic CSV
DevOps	GitHub Actions · Streamlit Cloud · pytest · Hypothesis

6 Deployed Features

Symptom Assessment Chatbot · Personal Health Dashboard · BC Facility Finder · Health Education Library · System Analytics Dashboard · Secure Authentication

Architecture Overview — CedarCare



Challenges Faced

01 CI/CD Module Path Conflicts

THE PROBLEM

Tests and source code were in different folders. GitHub Actions couldn't find source files when running tests — causing three consecutive CI pipeline failures.

THE FIX

Explicit PYTHONPATH and working directory set in the GitHub Actions YAML.

KEY LEARNING

Directory structure decisions made early have consequences that only surface at deployment.

02 Streamlit Re-Run Lifecycle

THE PROBLEM

Streamlit re-executes the entire script on every user interaction. Chip buttons appeared at wrong times and turn counters reset unexpectedly mid-conversation.

THE FIX

Dedicated `_show_chips` session state flag and careful turn counter management.

KEY LEARNING

Understanding a framework's execution model shapes every architectural decision you make.

03 GPT Prompt Engineering for JSON

THE PROBLEM

GPT-4o-mini misclassified "Moderate" as a duration instead of severity — because the word is ambiguous without context.

THE FIX

Explicit JSON schema enforcement in the system prompt and a severity-word blocklist.

KEY LEARNING

LLM outputs need the same defensive validation mindset as any external API.

Lessons Learned

01 You cannot design for users without talking to users

WHAT HAPPENED

User feedback led me to replace the original KPI cards with more meaningful metrics — and most importantly, it prompted me to introduce a proper landing page that users actually wanted to see first.

TAKEAWAY

Design assumptions are just guesses until someone outside the project tests them.

02 The best model on paper isn't always the best model in production

WHAT HAPPENED

SVM hit 96% accuracy but produced a 180MB model file — too large for Streamlit Cloud startup. Random Forest hit 100% accuracy and loaded in milliseconds. The benchmark made the right choice obvious.

TAKEAWAY

Don't just measure accuracy. Measure training time, model size, and deployment fit. Let the data decide.

03 Documenting limitations honestly is harder than building features

WHAT HAPPENED

SQLite resets on Streamlit Cloud restart. The Kaggle dataset is too clean for real-world use. Streamlit has UI ceilings. I documented all three clearly rather than hiding them.

TAKEAWAY

Knowing where your system breaks — and saying so clearly — is what separates a credible research project from one that merely functions.

 LIVE

Demo

cedarcare.streamlit.app

1

Landing Page

Unauthenticated experience · CedarCare branding · HTML/CSS via st.components

2

Login

bcrypt authentication · session_state management

3

Chatbot: Happy Path

UTI → severity chip → duration chip → Random Forest result (100% confidence)

4

Chatbot: Emergency

Severe chest pain, difficulty breathing → CTAS Level 1 fires · ML never runs

5

Chatbot: OpenAI Fallback

"I have a headache" → RF confidence <70% → OpenAI general guidance

6

Chatbot: Non-Medical Redirect

"What's the weather today?" → zero symptoms → redirect → hard stop turn 2

7

Personal Health Dashboard

KPI cards · Plotly charts · date range filter (fast)

8

System Analytics

K-Means cluster scatter · 3 patient segments (fast)

9

Facility Finder

Interactive Folium map · BC healthcare facilities (fast)

10

Health Library

18 articles · 8 categories · full-text search (fast)

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Thank you for your time.

Questions?

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LIVE DEPLOYMENT

cedarcare.streamlit.app

GITHUB REPOSITORY

[desporamon/W26_4495_S2_DesmondC](https://github.com/desporamon/W26_4495_S2_DesmondC)

TEST COVERAGE

95 tests · 79/79 CI passing

USER SURVEY

4.42/5 likelihood to recommend · 12 respondents

Product Evaluation

Three independent evaluation methods — algorithm benchmarking, automated testing, and user research.

METHOD 1

Algorithm Benchmarking

Algorithm	Accuracy	Model Size	Verdict
Random Forest	100%	Small	✓ Selected
SVM	96%	180 MB	✗ Too large
KNN	94%	Moderate	✗ Lower acc.

RF won on both accuracy and deployment fit. SVM's 180MB file would cause Streamlit Cloud timeouts on startup.

METHOD 2

Automated Testing Suite

Unit Tests	57
Integration Tests	12
UI Tests (AppTest)	16
Property-Based (Hypothesis)	10

95 total · 79/79 passing in CI
Hypothesis caught 2 bugs that example-based tests missed entirely.

METHOD 3

User Survey

12

Respondents

83%

Named chatbot most useful

4.33

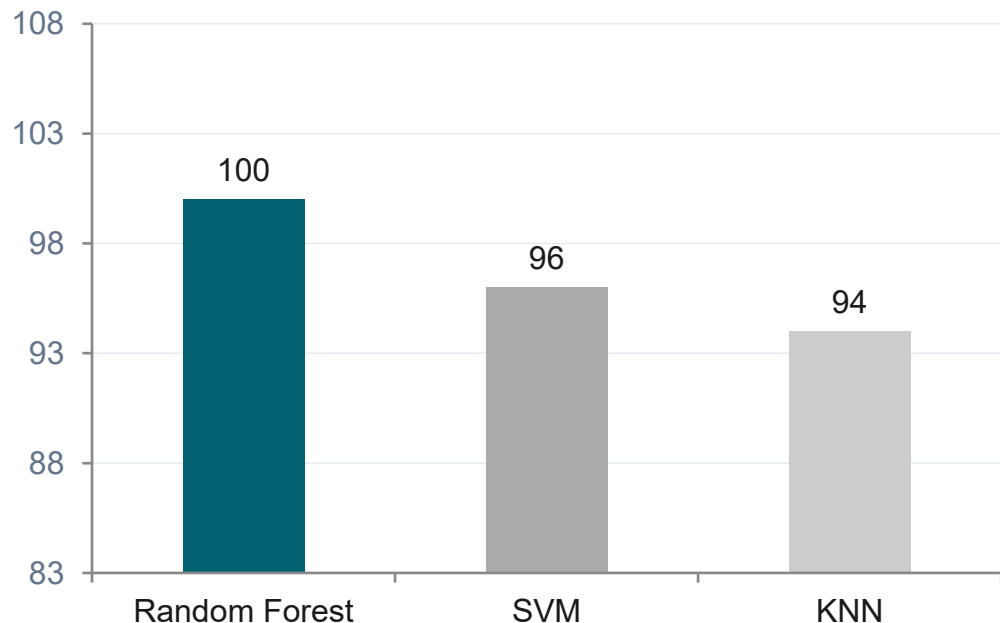
Ease of use /5

4.42

Likelihood to recommend /5

6-question Google Forms survey. Conducted after informal housemate testing and formal demo sessions.

ML Algorithm Benchmarking — Why Random Forest?



Algorithm	Accuracy	Train Time	Model Size	Verdict
Random Forest	100%	Fast	Small	✓ Selected
SVM	96%	45 seconds	180 MB	✗ Too large
KNN	94%	Moderate	Moderate	✗ Lower accuracy

Why accuracy alone wasn't enough

SVM produced a 180MB model — too large for Streamlit Cloud startup. RF loaded in milliseconds via joblib and cached with `st.cache_resource`. Accuracy + deployment fit = RF wins.

Note: 100% accuracy reflects the Kaggle dataset's clean, non-overlapping symptom-disease mappings. Real clinical data would yield lower accuracy — which is why the 70% confidence threshold and OpenAI fallback exist as safety mechanisms.

Testing Architecture — 95 Tests, 4 Methodologies

Test File	Type	Tests	Key Validated Behaviour	CI
test_auth.py	Unit	10	Registration, login, bcrypt hashing, session management, page protection	✓
test_ctas_rules.py	Unit	23	14 CTAS rule detections, 5 negative pass-throughs, 4 edge cases	✓
test_database.py	Unit	12	User CRUD, Assessment CRUD, data isolation, schema initialisation	✓
test_ml_model.py	Unit	12	Model loading, metadata, confidence threshold, input/output validation	✓
test_pipeline.py	Integration	6	Emergency circuit breaker, ML path, OpenAI fallback, precedence	✓
test_chatbot_scenarios.py	Integration	6	Cardiac emergency, stroke, ML (UTI), fallback, emergency override	✓
test_dashboard_apptest.py	UI (AppTest)	8	Dashboard page load, KPI rendering, filter application	Local
test_chatbot_apptest.py	UI (AppTest)	8	Chatbot page load, message submission, state management	Local
test_fuzz_hypothesis.py	Property-Based	10	Password/email/CTAS fuzzing; database corruption resistance	✓

95 total tests · 79/79 passing in GitHub Actions CI · 16 AppTest tests run locally · Hypothesis caught 2 bugs missed by example-based tests